GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

HYBRID GREEN KERNEL

$$G_\theta = \text{Dirac delta structure} + \text{Heaviside cancellation} + \text{learned smooth correction.}$$

DIRAC DELTA STRUCTURE

Provide the source singularity.

HEAVISIDE CANCELLATION

Cancel the Heaviside contribution generated by the Dirac-delta jump.

LEARNED SMOOTH CORRECTION

Learn the remaining smooth residual.

Built-in singular and cancellation structure reduces what the neural network must learn.

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$$G_\theta(t,\eta) = \underbrace{E(t,\eta)M(t)R_\theta(t,\eta)}_{\text{learned smooth correction}} + \underbrace{B(t)\left(J_0(t,\eta) - \frac{1}{2}E(t,\eta)\right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t,\eta)}_{\text{Dirac delta jump}}.$$

DIRAC DELTA STRUCTURE

$$G_0(t,\eta) = \begin{cases} t(1-\eta), & t<\eta, \\ \eta(1-t), & t\geq\eta, \end{cases} \qquad \partial_t^2 G_0 = -\delta(t-\eta).$$

HEAVISIDE CANCELLATION

$$\partial_t J_0(t,\eta) = G_0(t,\eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

COEFFICIENT FACTORS

$$A(t) = \frac{1}{a_{\text{unit}}(t)}, \qquad B(t) = \frac{a'_{\text{unit}}(t) + b_{\|,\text{unit}}(t)}{a_{\text{unit}}(t)^2}.$$

The analytic terms handle the Green-function singularity before learning.

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LEARNED SMOOTH CORRECTION

$$E(t,\eta)M(t)R_\theta(t,\eta), \qquad E(t,\eta) = t\eta(1-\eta), \qquad M(t) = 1-t.$$

GreenNet learns R_θ, while $E(t,\eta)M(t)$ makes the correction boundary-compatible. The singular and cancellation structure is built into the kernel form.

The neural network learns only the smooth residual, with boundary-compatible envelope factors.